

Notes: Chincó and Sammon (JFE, 2024)
The Passive Ownership Share Is Double What You Think It Is

Contents

1	Summary	2
2	Big picture	2
3	Institutional setting	3
4	Core measurement equation	3
5	Main results	3
6	Empirical interpretation	5
7	Contribution	5
8	Notes-created summary tables	6
9	Extracted figures, tables, and key equations	8

1 Summary

This paper asks a measurement question with major implications for asset pricing and market structure: how much of the U.S. stock market is actually held by passive investors? The usual benchmark is the index-fund ownership share reported by the Investment Company Institute (ICI). In 2021, index mutual funds and ETFs owned about 16% of the U.S. stock market. Chincio and Sammon argue that this number misses a large amount of passive ownership outside registered index funds, including internal indexing, direct indexing, and active funds that are effectively closet indexing.

The key idea is to infer passive assets under management from reconstitution-day trading volume. When a stock is added to or deleted from an index, passive investors tracking that index must rebalance. If the stock enters an index with weight w and reconstitution-day volume is V shares at price P , then the implied assets tracking the index are approximately

$$\widehat{AUM}_{i,n,t} = \frac{V_{n,t} \times P_{n,t}}{w_{i,n,t}}.$$

Each index change therefore gives a point estimate of how much money is tracking that index. Averaging across changes and summing across five major indexes - the S&P 500, S&P MidCap 400, Russell 1000, Russell 2000, and Nasdaq 100 - yields the paper's headline estimate.

The headline finding is that passive investors tracking these five indexes owned 33.5% of the U.S. stock market in 2021, more than twice the ICI index-fund ownership share of 16%. Using only volume from 4:00pm to 11:59pm, a more conservative proxy for reconstitution-related trading, still gives 24.6%. The authors emphasize that their estimate covers only five indexes; if they also included CRSP indexes heavily used by Vanguard, the 2021 estimate would rise further.

A central contribution is not only the level estimate, but also the evidence on how passive investors trade. Rebalancing volume is highly concentrated at the market close on reconstitution day. This suggests that passive investors - including non-index-fund passive investors - rebalance in a highly synchronized way, often through prearranged trades. Because these trades are arranged before they are executed, the associated price impact can occur before reconstitution day. This helps explain why modern index reconstitutions can generate enormous volume with limited same-day price pressure and why short-window demand elasticity estimates can look artificially large.

2 Big picture

- **Question.** How large is the total passive ownership share once passive investors outside index mutual funds and ETFs are included?
- **Measurement challenge.** Index funds disclose holdings, but internal indexers, direct indexers, and closet indexers are much harder to observe directly.
- **Solution.** Use reconstitution-day volume as a revealed-demand measure for the dollars tracking an index.
- **Main estimate.** Passive ownership of the U.S. stock market is 33.5% in 2021 using daily reconstitution volume, versus 16% for the ICI index-fund ownership share.
- **Trading insight.** Passive rebalancing is concentrated at the close and often prearranged, so price effects can occur before the observed demand shock.

3 Institutional setting

The paper focuses on five major U.S. equity indexes: the S&P 500, the S&P MidCap 400, the Russell 1000, the Russell 2000, and the Nasdaq 100. Each index has periodic or ad hoc reconstitution events in which stocks are added to or dropped from the index. Passive investors tracking the index must adjust their portfolios to match the new weights.

The Russell indexes are particularly useful because annual reconstitution produces many simultaneous additions, deletions, and migrations. The S&P 500 and Nasdaq 100 have fewer changes, but their constituents are large and the dollar volume associated with rebalancing can be large. The authors assemble data on index membership, index weights, prices, and trading volume from 2000 to 2021.

The paper’s identifying logic relies on a mechanical constraint: if a stock enters an index at a known weight, the total dollar demand from index-tracking investors is proportional to the amount of money tracking that index. Reconstitution-day volume therefore reveals the passive ownership share if the volume spike is primarily passive rebalancing.

4 Core measurement equation

For an addition, let $IndexWeight_{i,n,t_{Recon}}$ denote stock n ’s weight in index i on reconstitution day t_{Recon} . If passive investors tracking the index need to build the new position, the value of required purchases is

$$IndexWeight_{i,n,t_{Recon}} \times AUMIndexed_{i,t_{Recon}}.$$

If daily reconstitution volume measures the passive rebalancing demand, then

$$DailyVolume_{n,t_{Recon}} \times Price_{n,t_{Recon}} = IndexWeight_{i,n,t_{Recon}} \times AUMIndexed_{i,t_{Recon}}.$$

The implied assets tracking index i from stock n are therefore

$$\widehat{AUMIndexed}_{i,n,t_{Recon}} = \frac{DailyVolume_{n,t_{Recon}} \times Price_{n,t_{Recon}}}{IndexWeight_{i,n,t_{Recon}}}.$$

The paper applies the same idea to deletions and alternative volume measures. For each index-year, it averages across all relevant additions and deletions, then scales by total U.S. stock-market capitalization:

$$\%Indexed_i(y) = 100 \times \frac{\widehat{AUMIndexed}_i(y)}{TotalMarketCap(y)}.$$

The total passive ownership estimate is the sum of the five index-specific shares.

5 Main results

5.1 Passive ownership is much larger than the index-fund share

The paper’s central estimate is that passive investors tracking the five studied indexes owned 33.5% of the U.S. stock market in 2021. This is more than twice the 16% ICI index-fund ownership share. The difference is interpreted as ownership by non-index-fund passive investors, such as institutions managing internal index portfolios, direct indexers, and closet indexers.

The estimates are economically large by index. In 2021, Table 3 reports passive ownership shares of 16.80% for the S&P 500, 10.77% for the Russell 1000, 3.76% for the Nasdaq 100, 1.12% for the S&P MidCap 400, and 1.06% for the Russell 2000. These sum to 33.50%.

5.2 Alternative volume proxies give lower but still large estimates

The paper reports a range of estimates to address potential measurement concerns. In 2021, using all daily volume gives 33.50%. Subtracting normal daily volume gives 28.65%. Using only volume from 4:00pm to 11:59pm gives 24.62%. Using closing-price volume gives 22.53%. The most conservative closing-auction proxy gives 13.31%, close to the ICI index-fund share but still based only on five indexes.

The interpretation is that daily volume can overstate passive demand if it includes active trading, while narrow closing-window volume can understate passive demand if some prearranged trades print outside the strict window. The main conclusion does not depend on one exact proxy: even conservative estimates imply more passive ownership than standard index-fund statistics capture.

5.3 Cross-validation from same-day index changes

A useful internal check is that different stocks added to or dropped from the same index on the same day should imply the same total AUM tracking that index. The paper emphasizes that these estimates line up closely. The measurement error in Table 5 is small for most indexes, and the full sample average absolute error is about 1 percentage point. This supports the claim that the volume spikes scale with index weights in the precise way predicted by passive rebalancing.

5.4 Reconstitution-day volume is large and concentrated

The trading-volume evidence is central to the paper. Figure 7 shows that volume spikes sharply on reconstitution day. Figure 9 shows that a much larger share of daily volume is executed from 4:00pm to 11:59pm on reconstitution days than on normal days. For the full sample, the after-hours and closing-window share rises from about 7.0% on a typical pre-reconstitution day to 64.5% on reconstitution day.

This concentration is difficult to reconcile with a story in which active investors gradually arbitrage around index changes. It is consistent with many investors trying to minimize tracking error by executing at or near the official closing price.

5.5 Prearranged trades help explain price behavior

The paper argues that a large ecosystem of rebalancing facilitators helps passive investors trade at the close. Passive investors can prearrange trades with intermediaries, and these arrangements can shift price impact before the actual reconstitution-day execution. This explains why reconstitution-day price pressure is no longer as large as earlier literature might predict despite enormous rebalancing volume.

The Tesla addition to the S&P 500 illustrates what happens when prearranging trades is unusually hard. Tesla's addition was surprising, very large, and difficult for liquidity providers to handle. In

that case, both volume and price movements around reconstitution were much more extreme than usual.

6 Empirical interpretation

6.1 What counts as passive?

The paper counts holdings as passive if they are rebalanced mechanically at the same point in time as index-tracking demand. This definition is intentionally behavior-based. A portfolio can be held by an active manager and still count as passive in this measurement if the manager is effectively benchmarking or closet-indexing and rebalances at the close on reconstitution day.

This definition is broader than the index-fund universe. It is also the reason the paper’s estimate is larger than ICI statistics. The point is not that the ICI number is wrong; rather, the ICI number measures only a disclosed subset of passive investing.

6.2 Why use volume rather than holdings?

Holdings data reveal index funds and some mutual funds, but they are incomplete for institutions that internally index, separately managed accounts, and other investors outside the public fund universe. Reconstitution-day volume has the advantage that all investors who rebalance at the same mechanical time leave a trading footprint. Each addition or deletion produces an observable implication for indexed AUM.

6.3 Potential threats

The main concern is that reconstitution-day volume may include non-passive trading. The paper addresses this in several ways. First, the same-index same-day cross-validation shows that individual stock volume scales closely with price and index weight. Second, volume is concentrated at the close, when passive rebalancing should occur. Third, alternative proxies based on narrower time windows still imply a large passive share. Fourth, there is little evidence that active traders are delaying their own trades in exactly the right proportions to mimic passive rebalancing demand.

A separate concern is that daily volume may miss passive trades arranged before reconstitution. The paper’s interpretation is that prearranged trades can affect prices before the final execution, so narrow event windows around reconstitution day can understate the total price impact of passive demand.

7 Contribution

The paper contributes to three literatures.

First, it contributes to the measurement of passive investing. It shows that index-fund holdings are not a sufficient measure of total passive ownership. Passive ownership can be inferred from reconstitution-day revealed demand, producing a much larger estimate.

Second, it contributes to index-effects and price-pressure research. If passive investors prearrange

trades, price effects can appear before reconstitution day rather than at the close. This helps reconcile high reconstitution-day volume with smaller same-day abnormal returns.

Third, it contributes to demand-system asset pricing. Researchers estimating demand elasticities from short event windows may conclude that demand is extremely elastic. The paper argues that once the event window includes the period when trades are arranged and prices adjust, demand looks much less elastic.

8 Notes-created summary tables

8.1 Measurement logic

Step	Object measured	Economic interpretation
Index change	Stock added to or dropped from an index	Mechanical portfolio demand
Weight	New index weight of the stock	Fraction of indexed AUM that must be traded
Volume spike	Reconstitution-day volume	Revealed rebalancing demand
AUM estimate	Volume times price divided by index weight	Dollars tracking the index
Market share	Indexed AUM divided by total U.S. market cap	Passive ownership share

8.2 Headline numbers

Estimate	2021 value	Interpretation
ICI index-fund share	16.0%	Disclosed domestic equity index mutual funds and ETFs tracking any index
Daily-volume estimate	33.5%	Passive ownership inferred from reconstitution-day volume across five indexes
4:00pm-11:59pm estimate	24.6%	Conservative estimate using late-day rebalancing volume
Closing-auction estimate	13.3%	Very narrow proxy that likely misses some prearranged passive trading
Five-index estimate plus CRSP indexes	38.5%	Approximate adjustment if large CRSP index-tracking holdings are included

8.3 Mechanism map

Question	Empirical evidence	Interpretation
Is reconstitution volume passive?	Volume scales with price and index weight across same-day index changes	Mechanical demand from index tracking
Is rebalancing gradual?	Little extra volume in the days immediately before, huge spike at close	Investors wait for official reconstitution price
Are non-index-fund passive investors important?	Implied passive share exceeds index-fund holdings	Internal indexing and closet indexing matter
Why limited same-day price pressure?	Price effects can occur when trades are prearranged, before execution	Demand shock affects prices before reconstitution day
Why do elasticity estimates vary?	Longer horizons imply much less elastic demand than one-day windows	Short windows miss prearrangement effects

9 Extracted figures, tables, and key equations



The passive ownership share is double what you think it is[☆]

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ABSTRACT

Each time a stock gets added to or dropped from an index, we ask: "How much money would have to be tracking that index to explain the huge spike in rebalancing volume we observe on reconstitution day?" While index funds held 16% of the US stock market in 2021, we put the overall passive ownership share at 33.5%. Our headline number is twice as large because it reflects index funds as well as other kinds of passive investors, such as institutional investors with internally managed index portfolios and active managers who are closet

Title and abstract

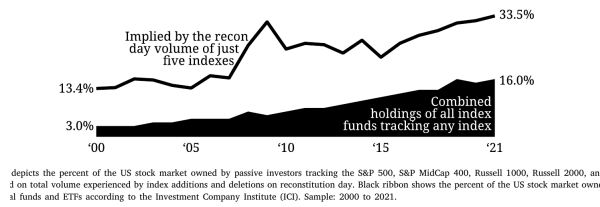


Figure 2: Passive share versus ICI index-fund share

Table 1

Characteristics of stocks that were added to or dropped from either the S&P 500, the S&P MidCap 400, the Russell 1000, the Russell 2000, or the Nasdaq 100. *MCap*: Market cap on reconstitution day in billions of dollars. *IndexWeight*: Weight in index in basis points. *ADV*: Average volume during the 22 trading days prior to reconstitution in millions of shares per day. *PastRet*: Return during the 6 months prior to reconstitution in percent. Sample: 2000 to 2021.

	Avg	Sd	Avg	Sd	Avg	Sd
Panel A: S&P 500						
	Adds			Drops		
<i>MCap</i> [\$1b]	12.7	25.1	14.6	31.8	10.9	15.3
<i>IndexWeight</i> [bps]	8.7	12.6	9.4	13.6	8.0	11.4
<i>ADV</i> [1 m]	4.9	11.1	3.3	5.8	6.4	14.3
<i>PastRet</i> [%]	10.5	41.4	21.7	41.7	-0.9	37.9
Panel B: S&P MidCap						
	Direct Adds			Direct Drops		
<i>MCap</i> [\$1b]	2.8	2.1	3.1	1.8	2.4	2.3
<i>IndexWeight</i> [bps]	19.5	13.8	22.2	12.2	16.1	14.9
<i>ADV</i> [1 m]	1.5	3.6	1.1	1.7	2.0	4.9
<i>PastRet</i> [%]	12.2	44.7	22.1	45.1	0.2	41.1
Panel C: Russell 1000						
	Direct Adds			Migrations		
<i>MCap</i> [\$1b]	4.4	6.9	8.9	12.8	3.0	2.1
<i>IndexWeight</i> [bps]	2.4	4.4	4.5	8.7	1.8	0.7
<i>ADV</i> [1 m]	1.8	5.0	2.8	5.7	1.4	4.7
<i>PastRet</i> [%]	27.2	62.7	18.1	60.2	29.9	63.2
Panel D: Russell 2000						
	Direct Adds			Migrations		
<i>MCap</i> [\$1b]	0.6	0.6	0.4	0.5	1.2	0.7
<i>IndexWeight</i> [bps]	3.5	3.5	2.4	2.4	8.5	3.8
<i>ADV</i> [1 m]	0.7	2.4	0.5	2.1	1.8	3.5
<i>PastRet</i> [%]	39.2	129.6	49.4	138.8	-8.6	48.7
Panel E: Nasdaq 100						
	Adds			Drops		
<i>MCap</i> [\$1b]	12.0	18.6	15.8	22.4	8.1	12.8
<i>IndexWeight</i> [bps]	29.8	39.3	38.3	43.0	21.4	33.2
<i>ADV</i> [1 m]	6.4	23.6	5.6	17.1	7.3	28.6
<i>PastRet</i> [%]	11.3	48.9	30.3	51.5	-7.6	37.7

Table 1: Stock characteristics

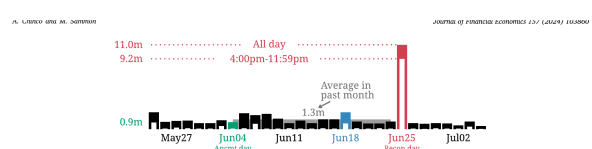


Figure 1: YETI volume around Russell reconstitution

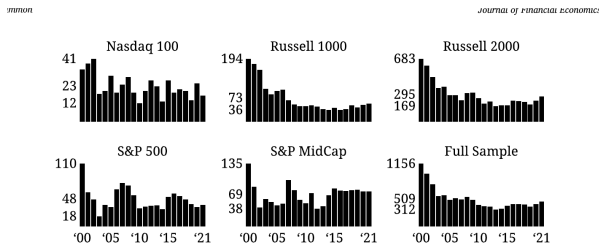


Figure 3: Number of index additions and deletions

Table 2

Reconstitution-day volume for stocks that were added to or dropped from either the S&P 500, the S&P MidCap 400, the Russell 1000, the Russell 2000, or the Nasdaq 100. *DailyVolume*: Total volume. *Volume1600to2359*: Volume from 4:00pm to 1:59pm. *VolumeAtClosingPrice*: Volume at the closing price. *VolumeAtClose*: Closing volume. All volume measures are normalized by *ADV*. Data for *Volume1600to2359*, *VolumeAtClosingPrice*, and *VolumeAtClose* start in 2004.

$\times ADV$	Avg	Sd	Avg	Sd	Avg	Sd
Panel A: S&P 500						
	Adds			Drops		
<i>DailyVolume</i>	12.4	9.9	16.5	10.6	8.4	7.2
<i>Volume1600to2359</i>	8.6	8.4	12.6	9.2	5.1	5.6
<i>VolumeAtClosingPrice</i>	6.5	9.4	9.0	8.1	4.3	9.8
<i>VolumeAtClose</i>	2.7	4.6	4.1	5.7	1.4	2.7
Panel B: S&P MidCap						
	Direct Adds			Direct Drops		
<i>DailyVolume</i>	10.8	8.4	12.5	8.4	9.0	8.1
<i>Volume1600to2359</i>	8.0	7.1	9.6	6.9	6.3	6.8
<i>VolumeAtClosingPrice</i>	5.7	5.7	6.5	5.7	4.9	5.7
<i>VolumeAtClose</i>	3.1	4.5	3.7	4.8	2.3	4.0
Panel C: Russell 1000						
	Direct Adds			Migrations		
<i>DailyVolume</i>	5.4	4.2	4.6	4.2	5.6	4.1
<i>Volume1600to2359</i>	4.6	3.7	2.5	1.9	5.1	3.8
<i>VolumeAtClosingPrice</i>	3.3	2.9	1.8	1.3	3.7	3.0
<i>VolumeAtClose</i>	1.4	2.2	0.8	1.2	1.5	2.3
Panel D: Russell 2000						
	Direct Adds			Migrations		
<i>DailyVolume</i>	14.2	17.1	15.9	18.3	6.3	4.9
<i>Volume1600to2359</i>	14.2	16.8	15.8	17.7	5.4	4.3
<i>VolumeAtClosingPrice</i>	9.8	20.5	10.9	22.0	4.1	4.8
<i>VolumeAtClose</i>	6.4	8.7	7.4	9.1	1.2	2.0
Panel E: Nasdaq 100						
	Adds			Drops		
<i>DailyVolume</i>	4.7	4.4	4.7	3.1	4.8	5.4
<i>Volume1600to2359</i>	2.4	2.3	2.3	1.5	2.6	2.9
<i>VolumeAtClosingPrice</i>	1.5	1.7	1.3	1.0	1.6	2.2
<i>VolumeAtClose</i>	1.5	1.6	1.5	1.1	1.6	2.0

Table 2: Reconstitution-day volume

5.1. Estimation strategy

Here is the core intuition behind our approach. Suppose stock ADD replaced stock DROP in index i at market close on day t_{Recon} . Let $AUM_{indexed,i}(t_{Recon})$ denote the total AUM held by passive investors tracking this index on reconstitution day. Further suppose that ADD initially represented $IndexWeight_{i,ADD}$ of the index. If passive investors perfectly matched this portfolio weight, then they had to build new positions worth

$$IndexWeight_{i,ADD} \times AUM_{indexed,i}(t_{Recon}) \quad (2)$$

Now imagine that passive investors are the only people trading ADD on reconstitution day and that these passive investors do all their trading at market close. In this scenario, $DailyVolume_{ADD}(t_{Recon})$ as reported in CRSP would capture all passive rebalancing volume. And these trades would be worth

$$DailyVolume_{ADD}(t_{Recon}) \times Price_{ADD}(t_{Recon}) \quad (3)$$

where $Price_{ADD}(t)$ denotes ADD's closing price per share on day t .

We impute the total AUM tracking index i by equating (2) and (3) and solving for $AUM_{indexed,i}(t_{Recon})$

$$AUM_{indexed,i,ADD}(t_{Recon}) \leftarrow \frac{DailyVolume_{ADD}(t_{Recon}) \times Price_{ADD}(t_{Recon})}{IndexWeight_{i,ADD}} \quad (4)$$

Key equations: implied indexed AUM

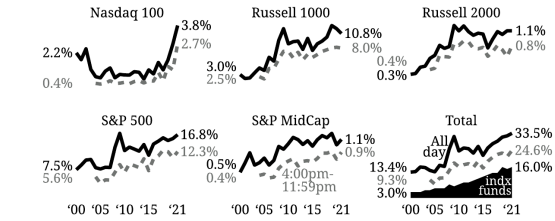


Figure 5: Percent of the US stock market owned by passive investors when using $DailyVolume$ as proxy for passive rebalancing: 2000 to 2021. B: funds according to the Investment Company Institute. Dotted line is percent owned by passive investors when using $Volume1600x2359$ 121.

Figure 5: Passive ownership by index

Table 4
Total passive ownership share as implied by six alternative proxies for passive rebalancing on reconstitution days. Column (1): $DailyVolume$. Column (2): $DailyVolume - ADV$. Column (3): $Volume1600x2359$. Column (4): $VolumeAtClosingPrice$. Column (5): $VolumeAtClose$. Numbers in parentheses are standard errors clustered by announcement. Sample: 2004 to 2021.

	All day (1)	Minus ADV (2)	4:00pm-11:59pm (3)	Closing price (4)	Closing auction (5)
2004	14.26 (1.42)	11.84 (1.52)	9.35 (0.47)	5.47 (0.40)	1.09 (0.21)
2005	13.52 (1.39)	11.36 (1.48)	7.18 (0.49)	4.40 (0.40)	0.97 (0.22)
2006	16.92 (1.59)	14.12 (1.39)	10.30 (0.75)	6.53 (0.50)	1.80 (0.30)
2007	16.33 (1.33)	13.37 (1.27)	10.41 (0.70)	7.42 (0.61)	1.21 (0.21)
2008	25.31 (1.80)	20.42 (1.50)	14.20 (0.80)	10.83 (0.70)	2.85 (0.30)
2009	31.85 (1.90)	27.64 (1.60)	18.15 (0.80)	9.91 (0.60)	4.92 (0.30)
2010	24.32 (1.80)	20.61 (1.50)	16.12 (0.80)	9.62 (0.60)	4.58 (0.30)
2011	25.90 (1.80)	21.86 (1.40)	17.21 (0.70)	10.78 (0.60)	3.81 (0.30)
2012	25.54 (1.80)	22.33 (1.40)	17.76 (0.70)	9.87 (0.60)	4.17 (0.30)
2013	23.21 (1.59)	19.51 (1.77)	15.82 (0.60)	10.31 (0.60)	3.54 (0.30)
2014	26.76 (1.80)	23.08 (1.60)	19.09 (0.80)	12.02 (0.80)	4.24 (0.30)
2015	22.06 (1.32)	19.92 (1.31)	15.47 (0.50)	11.86 (0.50)	5.52 (0.20)
2016	25.92 (1.50)	22.83 (1.30)	19.04 (0.70)	15.42 (0.70)	6.45 (0.30)
2017	28.11 (1.80)	24.67 (1.60)	21.09 (0.80)	17.46 (0.80)	9.99 (0.30)
2018	29.43 (1.80)	25.98 (1.60)	22.56 (0.80)	19.06 (0.80)	11.38 (0.30)
2019	31.53 (1.80)	25.12 (1.60)	23.29 (0.80)	21.71 (0.80)	10.79 (0.30)
2020	32.19 (1.80)	26.24 (1.60)	21.37 (0.70)	19.35 (0.70)	11.54 (0.30)
2021	33.50 (1.80)	28.65 (1.91)	24.62 (0.70)	22.53 (0.70)	13.31 (0.30)

Table 4: Alternative rebalancing-volume proxies

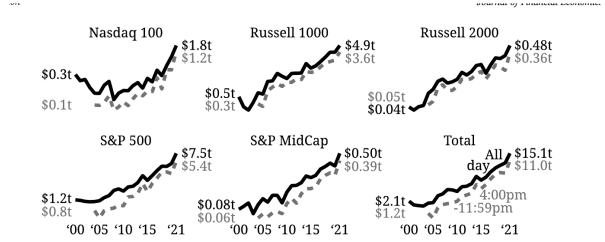


Figure 4: Implied passive AUM by index

Figure 4: Implied passive AUM by index

Table 3
Percent of the US stock market owned by passive investors when using $DailyVolume$ as proxy for passive rebalancing volume. Numbers in parentheses are standard errors clustered by announcement. Sample: 2000 to 2021.

	S&P 500	S&P MidCap	Russell 1000	Russell 2000	Nasdaq 100	Total
2000	7.50 (0.00)	0.49 (0.00)	2.96 (0.00)	0.28 (0.00)	2.16 (0.00)	13.39 (0.30)
2001	8.71 (0.00)	0.67 (0.00)	2.20 (0.00)	0.30 (0.00)	1.77 (0.00)	13.64 (0.30)
2002	10.07 (0.00)	0.86 (0.00)	2.30 (0.00)	0.41 (0.00)	2.40 (0.00)	16.04 (0.30)
2003	10.21 (0.00)	0.63 (0.00)	3.28 (0.00)	0.43 (0.00)	1.20 (0.00)	15.75 (0.30)
2004	8.03 (0.00)	0.67 (0.00)	4.21 (0.00)	0.57 (0.00)	0.78 (0.00)	14.26 (0.30)
2005	7.69 (0.00)	0.76 (0.00)	3.74 (0.00)	0.62 (0.00)	0.71 (0.00)	13.52 (0.30)
2006	8.14 (0.00)	0.63 (0.00)	6.26 (0.00)	0.80 (0.00)	1.10 (0.00)	16.92 (0.30)
2007	8.08 (0.00)	0.72 (0.00)	5.49 (0.00)	0.75 (0.00)	1.29 (0.00)	16.33 (0.30)
2008	13.40 (0.00)	0.98 (0.00)	9.44 (0.00)	0.78 (0.00)	0.71 (0.00)	25.31 (0.30)
2009	17.17 (0.00)	0.93 (0.00)	11.69 (0.00)	1.15 (0.00)	0.91 (0.00)	31.85 (0.30)
2010	12.56 (0.00)	1.04 (0.00)	8.79 (0.00)	1.06 (0.00)	0.88 (0.00)	24.32 (0.30)
2011	13.51 (0.00)	1.16 (0.00)	9.22 (0.00)	1.15 (0.00)	0.87 (0.00)	25.00 (0.30)
2012	12.98 (0.00)	1.11 (0.00)	9.33 (0.00)	1.04 (0.00)	1.07 (0.00)	25.54 (0.30)
2013	12.43 (0.00)	1.02 (0.00)	7.75 (0.00)	0.98 (0.00)	1.04 (0.00)	23.21 (0.30)
2014	14.18 (0.00)	1.09 (0.00)	9.83 (0.00)	1.02 (0.00)	0.64 (0.00)	26.76 (0.30)
2015	11.35 (0.00)	0.99 (0.00)	7.55 (0.00)	1.00 (0.00)	1.18 (0.00)	22.06 (0.30)
2016	14.32 (0.00)	1.11 (0.00)	8.80 (0.00)	0.75 (0.00)	0.94 (0.00)	25.92 (0.30)
2017	15.08 (0.00)	1.19 (0.00)	9.32 (0.00)	0.91 (0.00)	1.61 (0.00)	28.11 (0.30)
2018	15.97 (0.00)	1.15 (0.00)	10.09 (0.00)	1.05 (0.00)	1.17 (0.00)	29.43 (0.30)
2019	15.26 (0.00)	1.23 (0.00)	12.30 (0.00)	1.09 (0.00)	1.75 (0.00)	31.53 (0.30)
2020	15.77 (0.00)	1.01 (0.00)	11.68 (0.00)	1.06 (0.00)	2.67 (0.00)	32.19 (0.30)
2021	16.80 (0.00)	1.12 (0.00)	10.77 (0.00)	1.06 (0.00)	3.76 (0.00)	33.50 (0.30)

Table 3: Passive ownership shares by year

$$[0] \quad (2, 10) \quad (1, 96)$$

Table 5

Average absolute difference between the passive ownership share implied by change and the average passive ownership share implied by all changes to in the same year. Sample: 2000 to 2021.

	S&P 500	S&P MidCap	Russell 1000	Russell 2000	Nasdaq 100
[%Error]	5.49	0.35	1.87	0.23	0.74

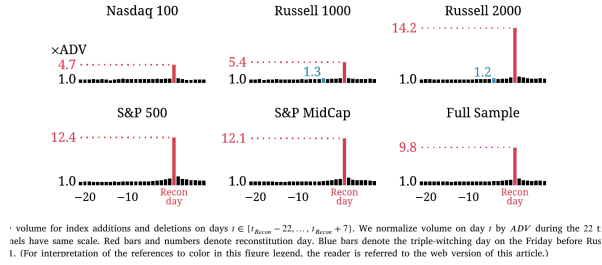
Table 5: Measurement-error magnitudes

Table 6

Predicting over- and underestimates in the passive ownership share. Each column reports the results of a separate univariate regression. The dependent variable is always the difference between the passive ownership share implied by each index change and the average passive ownership share implied by all changes to that index in the same year, $\%Error$. The right-hand-side variable is different in each column. $MCap$ is the market capitalization of the stock being added or dropped on the day before reconstitution in billions of dollars. Ret is the realized return of the stock being added or dropped on reconstitution day in percent. $SVolume$ is the dollar volume of the stock being added or dropped on reconstitution day in billions of dollars. $IndexWeight$ is the weight of the stock being added or dropped in basis points. $IsAddition$ is an indicator variable that is one if a stock is being added to the index and zero otherwise. $IsMigration$ is an indicator variable that is one if a stock is being moved between indexes and zero otherwise. Numbers in parentheses are t -stats clustered by announcement. Sample: 2000 to 2021.

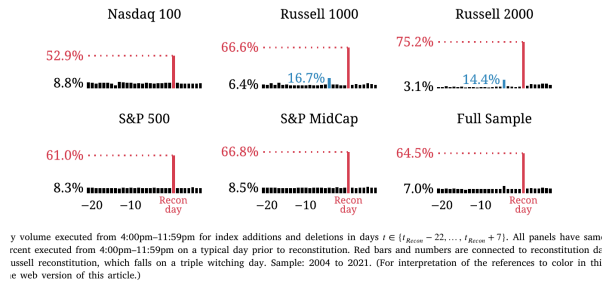
Dep variable:	(1)	(2)	(3)	(4)	(5)	(6)
Intercept	0.07** (2.26)	0.02 (0.89)	-0.06 (-1.20)	0.08*** (2.85)	-0.58*** (4.20)	-0.16*** (3.95)
$MCap$	-0.02*** (2.14)					
Ret		0.00 (0.06)				
$SVolume$			0.19 (1.58)			
$IndexWeight$				-0.01*** (4.04)		
$IsAddition$					0.67*** (6.71)	
$IsMigration$						0.65*** (5.71)
# Obs	11,168	11,087	11,168	11,168	11,168	11,168
Adj. R^2	0.66%	-0.01%	1.62%	0.28%	0.74%	1.16%

Table 6: Predicting over- and under-estimates



Volume for index additions and deletions on days $t \in [t_{Recon} - 22, \dots, t_{Recon} + 7]$. We normalize volume on day t by ADV during the 22 trading days prior to reconstitution. Red bars and numbers denote reconstitution day. Blue bars denote the triple-witching day on the Friday before Russell 2000 reconstitution. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

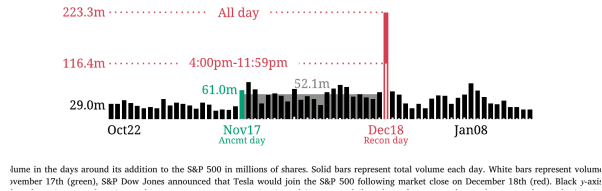
Figure 7: Daily volume around reconstitution



Volume executed from 4:00pm-11:59pm for index additions and deletions in days $t \in [t_{Recon} - 22, \dots, t_{Recon} + 7]$. All panels have same scale. Red bars and numbers are connected to reconstitution day. Russell reconstitution, which falls on a triple witching day. Sample: 2004 to 2021. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

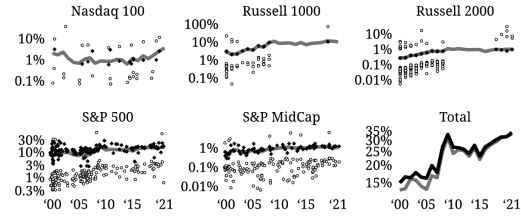
Figure 9: Late-day volume share

or this article.)



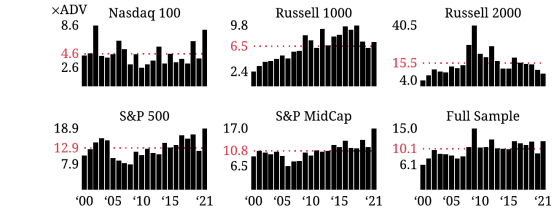
Volume in the days around its addition to the S&P 500 in millions of shares. Solid bars represent total volume each day. White bars represent volume on the day of the addition (Dec 18). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Figure 11: Tesla volume around S&P 500 addition



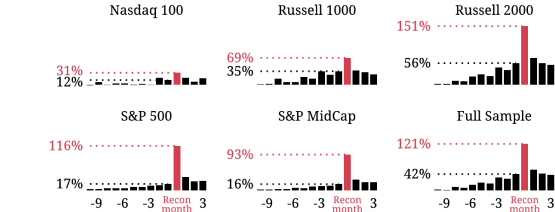
Events average passive ownership share tracking each index when using $DailyVolume$ as proxy for passive rebalancing volume. White dots represent the average passive ownership share implied by specific add/drop events which we classify as outliers. Black dots represent the average passive ownership share implied by the index on the same date. Black line is the total passive ownership share across all five indexes excluding outliers. Sample: 2000 to 2021.

Figure 6: Event-level estimates and outliers



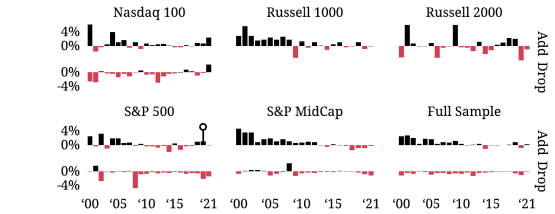
Note the average volume on reconstitution day experienced by index additions and deletions in a given year. We normalize each stock's volume during the 22 trading days prior to reconstitution. The highest and lowest y-axis labels in black represent the maximum and minimum volume on the reconstitution day. Red bars represent the time-series average over entire sample. Sample: 2000 to 2021. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Figure 8: Reconstitution-day volume by year



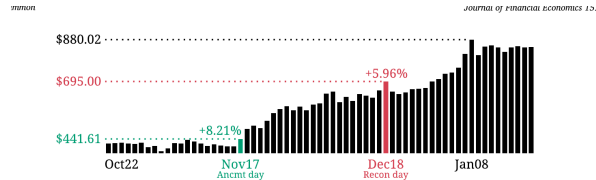
Volume for index additions and deletions in months $m \in [m_{Recon} - 10, \dots, m_{Recon} + 3]$. y-axis reports the percent difference between a stock's average daily volume 11 months prior to reconstitution, $100 \times \left(\frac{DailyVolume_{m+11}}{DailyVolume_{m-11}} - 1 \right)$. All panels have same scale. Red label denotes the month prior to reconstitution. Sample: 2001 to 2021. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Figure 10: Monthly volume around reconstitution



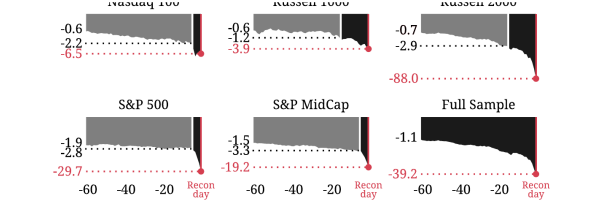
Note the average return in excess of the market on reconstitution day for a value-weighted portfolio of either index additions or index deletions. All panels have same scale. The bar for additions to the S&P 500 in 2020 does not include Tesla. The white circle shows the average return on the reconstitution day for the market for S&P 500 additions in 2020 when including Tesla. Sample: 2000 to 2021.

Figure 12: Reconstitution-day abnormal returns



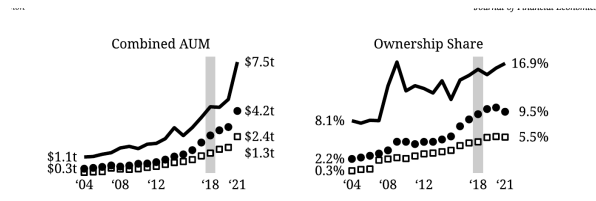
ing price in the days around its addition to the S&P 500. On November 17th (green), S&P Dow Jones announced that Tesla would join the S&P member 18th (red). Percentages reported in the figure are realized returns on announcement day and reconstitution day. Black y-axis label is h. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Figure 13: Tesla price around S&P 500 addition



and elasticity for direct additions at a particular horizon. The demand shock is reconstitution-day volume as a percent of its shares out excess return over the past $h \in \{60, \dots, 1\}$ trading days (x-axis). The red dot corresponds to the estimate at $h=1$, which only use a stock's left-most point in each panel uses a stock's price increase over the past quarter. The vertical white line in each panel denotes the average d reconstitution: 5 trading days for the S&P 500, S&P MidCap 400, and Nasdaq 100; 15 trading days for the Russell 1000 and Russell 2000. color in this figure legend, the reader is referred to the web version of this article.)

Figure 14: Demand elasticity by horizon



reflect mutual funds that increased their positions in S&P 500 additions and decreased their positions in S&P 500 deletions with probability squares reflect mutual funds that adjusted their holdings in the correct direction $\geq 80\%$ of the time. The solid line reflects the AUM trajectory. The vertical gray bar denotes 2018, the final year in Pavlova and Sikorskaya (2023).

Figure A.1: Fund holdings benchmark

Table B.1
Time-series regressions. Each column reports the results of a separate time-series regression using annual observations. Columns (1)-(6) include one observation per year. Column (7) is a pooled specification that includes five observations per year—i.e., one for each index. The state variable in panel A is our estimate for the passive ownership share. Sample: 2003 to 2021. The state variable in panel B is the change in these estimates for the passive ownership share. Lag # is the coefficient on a lagged value of the state variable. Year is the coefficient on the time trend. Numbers in parentheses are *t*-stats. Sample: 2004 to 2021.

Panel A: %Indexed							
	S&P 500 (1)	S&P MidCap (2)	Russell 1000 (3)	Russell 2000 (4)	Nasdaq 100 (5)	Total (6)	Pooled (7)
Intercept	8.58** (2.78)	0.38 (1.24)	2.99** (2.46)	0.29** (2.37)	-0.48 (1.08)	11.10** (2.23)	
Lag 1	0.36 (1.33)	0.33 (1.23)	0.41 (1.63)	0.48** (2.85)	0.67** (3.75)	0.45 (1.75)	0.58** (2.52)
Lag 2	-0.16 (0.73)	-0.03 (0.13)	0.13 (0.54)	0.11 (0.38)	0.18 (0.79)	-0.05 (0.23)	0.03 (0.16)
Lag 3	-0.20 (0.76)	0.21 (0.74)	-0.03 (0.13)	0.11 (0.45)	0.07 (0.31)	-0.11 (0.45)	0.06 (0.26)
Year	0.43** (3.34)	0.01 (0.50)	0.15 (1.07)	0.00 (0.05)	0.07** (2.04)	0.62** (3.01)	0.07** (2.44)
Index FE	N	N	N	N	N	N	Y
# Obs	19	19	19	19	19	19	95
R ²	68.6%	73.2%	71.1%	69.5%	70.3%	76.1%	97.5%

Panel B: $\Delta\%$ Indexed							
	S&P 500 (1)	S&P MidCap (2)	Russell 1000 (3)	Russell 2000 (4)	Nasdaq 100 (5)	Total (6)	Pooled (7)
Intercept	0.04 (0.03)	0.02 (0.05)	0.07 (0.35)	0.02 (0.06)	0.06 (0.43)	0.05 (0.20)	
Lag 1	-0.15 (0.50)	-0.19 (0.76)	-0.30 (0.78)	-0.17 (0.66)	-0.16 (0.70)	-0.20 (0.70)	-0.19* (1.65)
Lag 2	-0.24 (0.84)	-0.19 (0.63)	0.13 (0.40)	0.12 (0.57)	-0.05 (0.28)	-0.12 (0.48)	-0.02 (0.33)
Lag 3	-0.10 (0.73)	0.31 (1.00)	0.20 (0.67)	0.25 (0.93)	-0.15 (0.61)	0.08 (0.36)	0.01 (0.05)
Index FE	N	N	N	N	N	N	Y
# Obs	18	18	18	18	18	18	90
R ²	7.3%	31.9%	16.1%	10.2%	4.7%	5.9%	8.6%

Table B.1: Time-series regressions

Table B.1: Time-series regressions

Table B.2
Index-fund share of passive ownership. Coefficient estimates from regressing the index-fund ownership share on the overall passive ownership share. Each row represents a separate regression using annual observations. Panel A: Left-hand side is the ICI's index-fund ownership share, which reflects all index funds tracking any index. Right-hand side is the ownership share of passive investors tracking the Russell 1000, the Russell 2000, the S&P 500, the S&P MidCap 400, or the Nasdaq 100. Panel B: Left-hand side is ownership share of index funds tracking just these five indexes. Right-hand side is ownership share of passive investors tracking the same index. Numbers in parentheses are *t*-stats. Sample: 2000 to 2021.

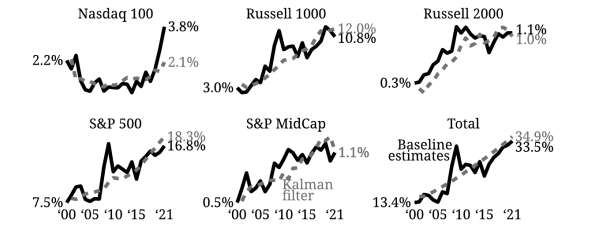
Panel A: ICI Factbook					
	Intercept	Slope	Adj. R ²	# Obs	
All index-funds that track any index	-4.01** (2.34)	0.54** (7.18)	70.7%	22	
Panel B: Thomson S12					
	Intercept	Slope	Adj. R ²	# Obs	
Funds that track the...					
S&P 500	-1.00** (0.33)	0.23** (2.75)	67.8%	22	
S&P MidCap	-0.18** (0.08)	0.42** (6.76)	67.9%	22	
Russell 1000	-0.09* (2.28)	0.04** (4.71)	74.0%	22	
Russell 2000	-0.06* (1.43)	0.25** (8.16)	78.2%	22	
Nasdaq 100	0.11** (3.14)	0.07** (3.14)	45.9%	22	
Funds that track one of these five indexes	-1.16** (2.43)	0.16** (5.32)	76.5%	22	

Table B.2: Index-fund share of passive ownership

Table C.1
Daily returns to the S&P 500 and a zero-cost strategy that rebalances the S&P 500's weights the day after S&P Dow Jones announces an upcoming reconstitution. Numbers in parentheses are standard errors. The first column reports results for the full sample: January 2nd 2002 through December 31st 2021. The next three columns report results for different subperiods: before Tesla's addition was announced; the day after the announcement through reconstitution day; and, the rest of the sample period.

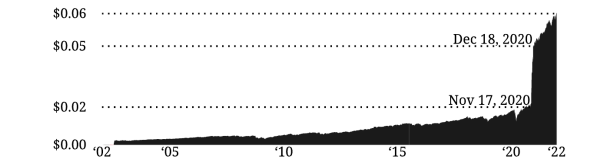
	Full Sample		Relative to Tesla's Addition	
	Start date	End date	Before	During
	Jan 2, 2002	Dec 31, 2021	Jan 2, 2002	Nov 17, 2020
S&P 500 (bps)	4.37** (0.73)	3.99** (0.75)	13.56 (39.06)	10.53 (26.08)
Lng/Short (bps)	0.02** (0.01)	0.01 (0.01)	2.67** (0.01)	0.01 (0.01)
# days	5,036	4,754	22	260

Table C.1: Returns to early rebalancing



es represent one-step-ahead forecasts produced by a Kalman filter. These filtered values represent our estimates for the passive ownership share.

Figure B.1: Kalman-filter comparison



returns to a zero-cost long/short strategy that starts off with \$1 on January 2nd 2002. The sample period ends December 31st 2021 rebalancing version of the S&P 500, which adjusts its weights to reflect each upcoming reconstitution the morning after the event is announced.

Figure C.1: Cumulative returns to early rebalancing

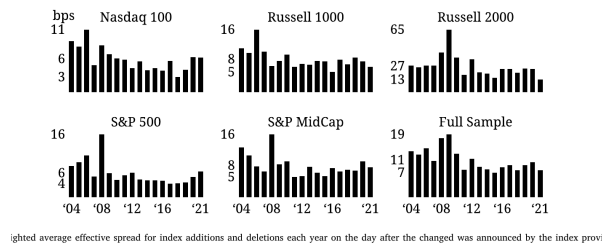


Figure C.2: Effective spreads after announcements

The ETF		Authorized Participant (AP)
Assets	Liabilities	
S&P 500	ETF shares	\$0

(a) When the market opens on June 12th, the ETF holds all 500 stocks in the S&P 500. Its position in Dish Network is worth \$3m.

The ETF		Authorized Participant (AP)
Assets	Liabilities	
S&P 500	ETF shares	\$0
+\$3m S&P 500	+\$3m ETF shrs	+\$3m ETF shrs

(b) At some point prior to the start of trading on June 16th, the AP creates \$3m worth of new ETF shares by delivering a \$3m position in S&P 500 stocks to the ETF.

The ETF		Authorized Participant (AP)
Assets	Liabilities	
S&P 500	ETF shares	\$0
+\$3m S&P 500	+\$3m ETF shrs	+\$3m ETF shrs
-\$3m DISH		+\$3m DISH

(c) Before markets close at 4:00pm on June 16th, the AP will arrange to redeem its \$3m in new ETF shares and receive the ETF's \$3m position in Dish Network at the close.

The ETF		Authorized Participant (AP)
Assets	Liabilities	
S&P 500	ETF shares	\$0
+\$3m S&P 500	+\$3m ETF shrs	+\$3m ETF shrs
-\$3m DISH		+\$3m DISH

(d) The AP uses either market-on-close (MOC) orders or pre-arranged upstairs trades to sell its newly acquired \$3m position in Dish Network. Because these trades are tied to the closing price, they get recorded at either 4:00pm (MOC orders) or in the minutes immediately after the close (upstairs trades).

Figure D.1: Mechanics of a prearranged rebalancing sale